Portfolio

or Notes on Introductory Quantum Optics by Gerry & Knight

FYST83

Author
Max Eriksson
maxerikss@gmail.com

Lund University
Department of Physics

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2 Field Quantization

2.1 Quantization of a Single Mode Field

For a cavity with ideal conducting walls, the electric field must vanish at the boundaries. This gives rise to standing waves. Maxwell's equations without generating sources are

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{B} = \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}, \quad \nabla \cdot \mathbf{B} = 0, \quad \text{and} \quad \nabla \cdot \mathbf{E} = 0. \tag{2.1}$$

Letting the cavity have the boundary positions $0, L$ and wave number $k = n\pi/L$ with $n \in \mathbb{N}$. The corresponding frequency is $\omega = c|k|$. Using Maxwell's equations we can find that

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{e}_x E(t) \sin(kz) \quad \text{and} \quad \mathbf{B}(\mathbf{r}, t) = \mathbf{e}_y B(t) \cos(kz). \tag{2.2}$$

If we define

$$E(t) = \omega \sqrt{\frac{2}{V\varepsilon_0}} q(t) \quad \text{and} \quad B(t) = \frac{1}{c} \sqrt{\frac{2}{V\varepsilon_0}} p(t). \tag{2.3}$$

The Hamiltonian is

$$H = \frac{1}{2}(p(t)^2 + \omega^2 q(t)^2) \tag{2.4}$$

With this we can verify that the expression for the Hamiltonian follows Hamilton's equations. That is, p, q are analogues to particle position and momentum. Is this a first sign of field quantizations and particles being the same? We can use the correspondence rule to replace the variables p, q with their operator counterparts $\hat{p}, \hat{q}$, which follow the canonical commutation relation $[\hat{p}, \hat{q}] = i\hbar$. This gives the operator form for the Hamiltonian

$$\hat{H} = \frac{1}{2}(\hat{p}^2 + \omega^2 \hat{q}^2). \tag{2.5}$$

Introducing creation and annihilation operators

$$\hat{a} = \frac{1}{\sqrt{2\hbar\omega}}(\omega\hat{q} + i\hat{p}) \quad \text{and} \quad \hat{a}^\dagger = \frac{1}{\sqrt{2\hbar\omega}}(\omega\hat{q} - i\hat{p}), \tag{2.6}$$

with the commutation relation $\left[\hat{a}, \hat{a}^\dagger\right] = 1$, which gives the Hamiltonian

$$\hat{H} = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2}\right), \tag{2.7}$$

showing that field excitations happen in units of $\hbar\omega$. For a given energy eigenstate $|n\rangle$ with energy E_n, the creation and annihilation operators have the properties

$$\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle \quad \text{and} \quad \hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle\,, \tag{2.8}$$

which means that for the energy $\hat{a}^\dagger \hat{H}\,|n\rangle = (E_n + \hbar\omega)(\hat{a}^\dagger\,|n\rangle)$, and similarly for $\hat{a}$ but with $-\hbar\omega$. Thus, we can see that these operators creates new eigenstates of the Hamiltonian with energy differing by $\hbar\omega$. Worth noting is that we are in the Schrödinger picture and the time-dependency is placed on the wave functions and not the operators. Furthermore, we can define the number operator $\hat{n} = \hat{a}^\dagger \hat{a}$ with the property $\hat{n}\,|n\rangle = n\,|n\rangle$. Which gives $E_n = \hbar\omega(n + 1/2)$. Also, the energy eigenstates $|n\rangle$ are orthogonal and form a complete set.

2.2 Quantum Fluctuations of a Single Mode Field

The electric and magnetic field operators are

$$\hat{E}(z, t) = \mathcal{E}_0(\hat{a} + \hat{a}^\dagger) \sin kz \quad \text{and} \quad \hat{B}(z, t) = -i\mathcal{B}_0(\hat{a} - \hat{a}^\dagger) \cos kz. \tag{2.9}$$

The field is not well defined for the number states $|n\rangle$ since $\langle n|\,\hat{E}\,|n\rangle = 0$, i.e. the mean field is zero, or we can see it as the field having a random phase. The square of the field, contributing to energy, is non-zero. We characterize fluctuations of a operator (field) by the standard deviation

$$\Delta\hat{O} = \sqrt{\left\langle \hat{O}^2 \right\rangle - \left\langle \hat{O} \right\rangle^2}. \tag{2.10}$$

Interestingly, $\Delta\hat{E}$ is non-zero even for $n = 0$, which means the field has vacuum fluctuations. Are these the fluctuations that can create particle-antiparticle pairs?

2.3 Quadrature Operators for a Single Mode Field

Defining quadrature operators, which can be considered dimensionless position and momentum operators,

$$\hat{X}_1 = \frac{1}{2}(\hat{a} + \hat{a}^\dagger) \quad \text{and} \quad \hat{X}_2 = \frac{1}{2i}(\hat{a} - \hat{a}^\dagger). \tag{2.11}$$

They have the commutation relation $\left[\hat{X}_1, \hat{X}_2\right] = i/2$ and thus $\Delta\hat{X}_1\Delta\hat{X}_2 \geq 1/4$, which is minimized for the vacuum state $|0\rangle$. According to field theory, the excitations, colloquilly called photons, are spread out over the entire mode, and are not localized. Doesn't this mean that the wave-particle duality is removed? And that instead of particles the wave itself is quantized in its properties. Or do we mean that the 'photon' is say a Gaussian wave packet of some sort?

2.4 Multimode Fields

To generalise to free space we model it as a cubic cavity with side length L where $L \to \infty$. We impose periodic boundary conditions on all faces, such that, $e^{ik_\nu \nu} = e^{ik_\nu(\nu+L)}$ with $k_\nu = (2\pi/L)n_\nu$ and $n_\nu \in \mathbb{Z}$ with $\nu \in \{x, y, z\}$. Considering spin, s, we obtain the Hamiltonian

$$\hat{H} = \sum_{\mathbf{k}s} \hbar\omega_k \left(\hat{a}^\dagger_{\mathbf{k}s}\hat{a}_{\mathbf{k}s} + \frac{1}{2}\right) \tag{2.12}$$

and $\hat{a}_{\mathbf{k}s}, \hat{a}^\dagger_{\mathbf{k}s}$ are defined similarly as in the single mode case. The multimode field operators are

$$\hat{\mathbf{E}}(\mathbf{r}, t) = i\sum_{\mathbf{k}s} A_k \mathbf{e}_{\mathbf{k}s}\left[\hat{a}_{\mathbf{k}s}e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_k t)} - \hat{a}^\dagger_{\mathbf{k}s}e^{-i(\mathbf{k}\cdot r-\omega_k t)}\right], \tag{2.13}$$

$$\hat{\mathbf{B}}(\mathbf{r}, t) = \frac{i}{c}\sum_{\mathbf{k}s}(\boldsymbol{\kappa} \times \mathbf{e}_{\mathbf{k}s})A_k\left[\hat{a}_{\mathbf{k}s}e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_k t)} - \hat{a}^\dagger_{\mathbf{k}s}e^{-i(\mathbf{k}\cdot r-\omega_k t)}\right], \tag{2.14}$$

where $\boldsymbol{\kappa} = \mathbf{k}/|\mathbf{k}|$ and $A_k = \sqrt{\hbar\omega_k/(2\varepsilon_0 V)}$. If $1/|\mathbf{k}| \gg |\mathbf{r}_{\text{atom}}|$, we can use the dipole approximation replacing the exponential $e^{i\mathbf{k}\cdot\mathbf{r}}$ with unity, optical radiation has a large enought wavelength to allow this. What are the limits of the dipole approximation? And what problems arises when we use it for systems where it is not valid.

2.5 Thermal Fields

A black body can be modelled as a cavity and assuming a weak coupling to an external heat bath, such that the system may be treated as isolated in the microcanonical ensemble, we can use the Hamiltonian from previous sections to describe the system. This gives us that the mean photon number, or the mean number of excitation in the field $\bar{n}$, is given by

$$\bar{n} = \frac{1}{e^{\hbar\omega/k_B T} - 1} \quad \text{and} \quad \bar{n} \approx \begin{cases} k_B T/\hbar\omega & k_B T \gg \hbar\omega \\ e^{-\hbar\omega/k_B T} & k_B T \ll \hbar\omega \end{cases} \tag{2.15}$$

The probability of finding n photons in the cavity is described by $P_n = \bar{n}^n/(1+\bar{n})^{n+1}$, that is $\hat{\rho}_{th} = \sum_n P_n |n\rangle\langle n|$. Interestingly, we also have that the fluctuations of the number operator are larger than the average photon number. Does this have any implications? Using the average energy per photon $\hbar\omega\bar{n}$ and calculating the energy per unit frequency and unit volume, using density of modes, we obtain Planck's law

$$\bar{U}(\omega, T) = \frac{\hbar\omega^3}{\pi^2 c^3}\frac{1}{e^{\hbar\omega/k_B T} - 1}. \tag{2.16}$$

Using the approximations in Eq. (2.15), we can derive Rayleigh's law ($k_B T \gg \hbar\omega$) and Wien's law ($k_B T \ll \hbar\omega$). Stefan-Boltzmann's law is obtained by integrating $\bar{U}(\omega, T)$ over all frequencies.

3 Coherent States

Coherent states are quantum states of the quantized electromagnetic field that gives a sensible classical limit, and can therefore be considered the most classical quantum states.

3.1 Eigenstates of the Annihilation Operator and Minimum Uncertainty States

The coherent states are defined as eigenstates of the annihilation operator such that $\hat{a}\,|\alpha\rangle = \alpha\,|\alpha\rangle$ for $\alpha \in \mathbb{C}$. Then it follows that $\langle\alpha|\,\hat{a}^\dagger = \langle\alpha|\,\alpha^*$. Expanding the coherent states in the number basis $|n\rangle$ and normalizing, we find

$$|\alpha\rangle = e^{-|\alpha|^2/2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}\,|n\rangle\,. \tag{3.1}$$

Theses states are minimum uncertainty states, satisfying equality in the Heisenberg uncertainty relation. It is possible to define the coherent state from this instead, minimizing the uncertainty for the quadrature operators. We also find that the expectation value of the field does not vanish, as it did for the number states, and the fluctuations is the same as for vacuum. From the definition of the coherent states and the number operator we find that $\bar{n} = |\alpha|^2$, and so α is related to the photon number. The fluctations of the number of photons in a coherent state follows a Poisson distribution, such that $\Delta n = \sqrt{\bar{n}}$ and the probability of finding n photons is given by $P_n = e^{-\bar{n}}\bar{n}^n/n!$. The fluctuations in the photon number thus increases with increasing photon number, and the relative fluctuations decreases as $\Delta n/\bar{n} = 1/\sqrt{\bar{n}}$. Furthermore, the coherent states are well localised in phase for large $\bar{n}$.

3.2 Displaced Vacuum States

We can also define the coherent states from the displacement operator acting on the vacuum state, such that $|\alpha\rangle = \hat{D}(\alpha)\,|0\rangle$, where $\hat{D}(\alpha) = \exp\big(\alpha\hat{a}^\dagger - \alpha^*\hat{a}\big) = \exp(-|\alpha|^2/2)\exp\big(\alpha\hat{a}^\dagger\big)\exp(-\alpha^*\hat{a})$ is a unitary operator. This operator also has the property that up to a global phase factor, $\hat{D}(\alpha)\hat{D}(\beta) = \hat{D}(\alpha+\beta)$, and the global phase factor does not affect the physics. Another property of the displacement operator is $\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) = \hat{a} + \alpha$, and $\hat{D}(\alpha)\hat{a}\hat{D}^\dagger(\alpha) = \hat{a} - \alpha$ The matrix elements of the displacement operator in the number state basis is given by

$$\langle m|\,\hat{D}(\alpha)\,|n\rangle = \sqrt{m!n!}e^{-|\alpha|^2/2}\sum_{l=0}^{\min(m,n)}\frac{\alpha^{m-l}(-\alpha^*)^{n-l}}{l!(m-l)!(n-l)!} \tag{3.2}$$

3.3 Wave Packets and Time Evolution

The time evolution of a coherent state can be described as $|\alpha, t\rangle = \exp(-i\omega t/2)\,|\alpha\exp(-i\omega t)\rangle$. The wave function for a coherent state in the position basis is described by a Guassian wave packet. The time evolution of this state is given by

$$\psi_\alpha(x,t) = \left(\frac{\omega}{\pi\hbar}\right)^{1/4}\exp\Big(-|\alpha|^2/2\Big)\exp\left(\frac{x^2\omega^2}{2\hbar^2}\right)\exp\left[-\left(\frac{x^2\omega^2}{\hbar^2} - \frac{\alpha e^{i\omega t}}{\sqrt{2}}\right)^2\right] \tag{3.3}$$

The integer spacing of the energy levels of the quantum harmonic oscillator gives rise to a Gaussian wave packet that does not change with time but rather oscillates back and forth, with the centroid follwoing a classical point mass in a harmonic potential. How do we use this property for trapped ions?

3.4 Generation of Coherent States

It is possible to generate coherent states by letting a classical oscillating current interact with the quantized vector field. The interaction energy may then be written as

$$\hat{V}(t) = -\sqrt{\frac{\hbar}{2\omega\varepsilon_0 V}}\left[\hat{a}\mathbf{e}\cdot\mathbf{J}(\mathbf{k},t)e^{-i\omega t} + \hat{a}^\dagger\mathbf{e}\cdot\mathbf{J}^*(\mathbf{k},t)e^{i\omega t}\right], \tag{3.4}$$

by introducing

$$u(t) = i\sqrt{\frac{1}{2\hbar\omega\varepsilon_0 V}}\mathbf{e}\cdot\mathbf{J}^*(\mathbf{k},t)e^{i\omega t} \quad \text{and} \quad \hat{V}(t) = i\hbar\left[u(t)\hat{a}^\dagger - u^*(t)\hat{a}\right] \tag{3.5}$$

we can write the time evolution operator for a finite time difference δt in the interaction picture as

$$\hat{U}(t+\delta t, t) = \exp\left(-\frac{i}{\hbar}\hat{V}(t)\delta t\right) = \exp\big(\delta t[u(t)\hat{a}^\dagger - u^*(t)\hat{a}]\big) = \hat{D}(u(t)\delta t), \tag{3.6}$$

where in the last step we identify the displacement operator and that δt is real. Taking the limit $\delta t \to 0$ while integrating over a finite time interval with the time ordering operator $\hat{\mathcal{T}}$, and ignoring the global phase factor we find that

$$\hat{U}(T,0) = \lim_{\delta t \to 0} \hat{\mathcal{T}} \prod_{n=0}^{T/\delta t} \hat{D}[u(t_n)\delta t] = \lim_{\delta t \to 0} \hat{D}\left[\sum_{n=0}^{T/\delta t} u(t_n)\delta t\right] = \hat{D}[\alpha(T)] \quad \text{and} \quad \alpha(T) = \int_0^T u(t)dt. \tag{3.7}$$

If the initial state is the vacuum state, the final state after the interaction, described by a displacment operator, is thus a coherent state $|\alpha(T)\rangle$, and $\alpha(T)$ is related to the classical current $\mathbf{J}(\mathbf{k},t)$. It is straightforward to show that $U(t,t') = \hat{D}[\alpha(t) - \alpha(t')]$ up to a global phase difference. Thus, the interpretation of this is that the time evolution of the vacuum state from t' to t is equivalent to the same displacement as the evolution from a coherent state $|\alpha(t')\rangle$ to $|\alpha(t)\rangle$.

3.5 More on the Properties of Coherent States

Unlike the number states, the coherent states are not orthogonal, and $\langle\alpha|\beta\rangle = \exp(-|\alpha-\beta|^2)$. They are however complete, satisfying the completeness relation

$$\int d^2\alpha\, |\alpha\rangle\langle\alpha| = \pi\mathbb{I} \quad \text{with} \quad d^2\alpha = d\Re\{\alpha\}\, d\Im\{\alpha\}\,. \tag{3.8}$$

The coherent states are thus overcomplete, and we can expand any state $|\psi\rangle$ in terms of coherent states by inserting the completeness relation. Furthermore, for an operator $\hat{F} = F(\hat{a},\hat{a}^\dagger)$, the diaganoal elements $\langle\alpha|\hat{F}|\alpha\rangle$ is sufficient to determine the operator completely in the number basis.

3.6 Phase-Space Pictures of Coherent States

Since canonical variables cannot be represented by a single point in phase.space due to them not commuting, they are instead represented by a distribution. The coherent states are minimum uncertainty states, with equal uncertainty in both quadratures, and expectation values $\langle\hat{X}_1\rangle_\alpha = \Re\{\alpha\}$ and $\langle\hat{X}_2\rangle_\alpha = \Im\{\alpha\}$, yielding coordinates in phase-space. The area of uncertainty is given by the circle with radius $\Delta\hat{X}_1 = \Delta\hat{X}_2 = 1/2$. Writing $\alpha = |\alpha|\exp(i\vartheta)$, we can therefore define any state $|\alpha\rangle$ by the magnitude and angle in phase-space. It is clear that for as $|\alpha|$ increases the uncertainty $\Delta\vartheta$ decreases. Comparing to number states, which have random phase but well-defined photon number, the phase-space representation is a circle around the origin with radius n. The time evolution of a coherent state in phase-space is a circular motion around the origin with angular frequency ω. When taking the expectation value for the electric field for example, we find that the result is a projection of this motion onto the $\hat{X}_1$-axis, yielding an oscillating field, with uncertainty given by the radius of the uncertainty circle. For the magnetic field we instead project onto the $\hat{X}_2$-axis.

3.7 Density Operators and Phase-Space Probability Distributions

Using coherent states we can write the density operator for a general state as

$$\hat{\rho} = \int d^2\alpha P(\alpha)\, |\alpha\rangle\langle\alpha|\,, \tag{3.9}$$

where $P(\alpha)$ is the Glauber-Sudarshan P-function, which is a quasiprobability distribution in phase-space, and can take negative values. States corresponding to $P(\alpha) \leq 0$ or states that are highly singular, meaning more singular than the delta function, are called nonclassical states. How something is more singular than a delta function is not clear to me. It requires defined derivatives of the function you integrate together with the derivative of a delta function, but why does this make it more singular?

Normal ordering of operators is defined such that all annihilation operators are to the right of all creation operators, anti-normal ordering is the opposite. For a normal ordered operator $:G(\hat{a},\hat{a}^\dagger):= G^{(N)}(\hat{a},\hat{a}^\dagger)$, the optical equivalence theorem states that the expectation value is given by

$$\left\langle G^{(N)}(\hat{a},\hat{a}^\dagger)\right\rangle = \int d^2\alpha P(\alpha) G^{(N)}(\alpha,\alpha^*) \tag{3.10}$$

which is thus a classical weighted average over phase-space with $P(\alpha)$ as weight.

Another distribution function is the Husimi Q-function, defined as

$$Q(\alpha) = \frac{\langle\alpha|\,\hat{\rho}\,|\alpha\rangle}{\pi}. \tag{3.11}$$

We can also calculate the expectation value of an operator using the Q- function and the Q-function is always positive. Lastly, we have the Wigner function, defined as

$$F_W(q,p) = \frac{1}{2\pi\hbar}\int_{-\infty}^{\infty} dx \left\langle q + \frac{1}{2}x \middle| \hat{\rho} \middle| q - \frac{1}{2}x \right\rangle e^{-ipx/\hbar}. \tag{3.12}$$

The Wigner function is a quasiprobability distribution, with negative values. Similarly to the P- and Q-functions, we can calculate expectation values of operators. However, for the Wigner function we have to use Weyl (symmetric) ordering of the operators, for example $\{\hat{p}\hat{q}\}_W = (\hat{p}\hat{q} + \hat{q}\hat{p})/2$.

4 Emission and Absorption of Radiation by Atoms

4.1 Atom-Field Interactions

An electron bound to an atom in an external field, can be described by the Hamiltonian

$$\hat{H}(\mathbf{r},t) = \frac{1}{2m}\left[\hat{\mathbf{P}} + e\mathbf{A}(\hat{\mathbf{r}},t)\right]^2 + e\Phi(\hat{\mathbf{r}},t) + V(\hat{\mathbf{r}}), \tag{4.1}$$

where the momentum operator $\hat{\mathbf{P}} = -i\hbar\nabla$, V is the Coulomb potential, and Φ,$\mathbf{A}$ are the scalar and vector potentials of the external field. The fields are given by

$$\mathbf{E}(\mathbf{r},t) = -\nabla\Phi(\mathbf{r},t) - \frac{\partial\mathbf{A}(\mathbf{r},t)}{\partial t} \quad \text{and} \quad \mathbf{B}(\mathbf{r},t) = \nabla\times\mathbf{A}(\mathbf{r},t), \tag{4.2}$$

invariant under gauge transformations $\Phi' = \Phi - \partial_t\chi(\mathbf{r},t)$ and $\mathbf{A}' = \mathbf{A} + \nabla\chi(\mathbf{r},t)$. We choose the Coulomb gauge $\nabla\cdot\mathbf{A} = 0$ gives $\Phi = 0$, and since an atom is much smaller than the wavelength of the field $\mathbf{k}\cdot\mathbf{r} \approx 0$. We place the atom at the origin and neglect the spatial dependence, giving $\mathbf{A}(\mathbf{r},t) \simeq \mathbf{A}(t)$ and $\Phi(\mathbf{r},t) \simeq \Phi(t)$. Performing the gauge transformation $\chi(\mathbf{r},t) = -\mathbf{A}(t)\cdot\mathbf{r}$ gives the Hamiltonian

$$\hat{H}(\mathbf{r},t) = \frac{\hat{\mathbf{P}}^2}{2m} + V(\hat{\mathbf{r}}) - \hat{\mathbf{d}}\cdot\mathbf{E}(t) = \hat{H}_0 - \hat{\mathbf{d}}\cdot\mathbf{E}(t), \tag{4.3}$$

where $\hat{\mathbf{d}} = -e\hat{\mathbf{r}}$ is the dipole moment. This is called Göppert-Mayer gauge, and contains one interaction term with the field, the electic dipole interaction. The electric field may be either classical or quantized. It would be interesting to see if other choices of gauge and gauge transformations could be used, and also test the validity of the dipole approximations for larger systems.

4.2 Interaction of an Atom with a Classical Field

We perform time-dependant perturbation theory, assuming $H_I = -\hat{\mathbf{d}}\cdot\mathbf{E}(t)$, is small. Assuming that the system is initially in a state $|i\rangle$ and due to the interaction it transition to a state $|f\rangle$. The probability of being in the final state is

$$P_f(t) = \frac{1}{\hbar^2}\left|\langle f|\,\hat{\mathbf{d}}\cdot\mathbf{E}_0\,|i\rangle\right|^2 \frac{\sin(\Delta t/2)}{\Delta^2} \overset{\Delta\to 0}{=} \frac{1}{\hbar^2}\left|\langle f|\,\hat{\mathbf{d}}\cdot\mathbf{E}_0\,|i\rangle\right|^2 t^2, \tag{4.4}$$

where $\Delta = \omega - \omega_{fi}$ is the detuning between the field frequency and the transition frequency. The transition probability rate is given by Fermi's golden rule

$$W_{i\to f} \approx \frac{2\pi}{\hbar}\left|\langle f|\,\hat{\mathbf{d}}\cdot\mathbf{E}\,|i\rangle\right|^2 \delta(\omega - \omega_{fi}), \tag{4.5}$$

where $\hbar\omega_{fi}$ is the energy difference between the initial and final state, and ω is the frequency of the field. This is generally accurate for incoherent light, for coherent/laser light, we need to consider the Rabi model.

4.3 Interaction of an Atom with a Quantized Field

The interaction Hamiltonian for a quantized field with the dipole approximation in Schrödinger picture is given by $\hat{H}_I = -\hat{\mathbf{d}} \cdot \sum_{\mathbf{k}s} \mathbf{E}_{0\mathbf{k}s}(\hat{a}_{\mathbf{k}s} - \hat{a}^\dagger_{\mathbf{k}s})$. We also consider the free-field Hamiltonian $\hat{H}_F = \sum_{\mathbf{k}s} \hbar\omega_{\mathbf{k}s}(\hat{a}^\dagger_{\mathbf{k}s}\hat{a}_{\mathbf{k}s} + 1/2)$, we count modes as j with $\mathbf{k}_j, s_j$. The total Hamiltonian is then $\hat{H} = \hat{H}_A + \hat{H}_F + \hat{H}_I = \hat{H}_0 + \hat{H}_I$. We have the inital state $|i\rangle = |\phi_i\rangle \otimes \bigotimes_j |n_j\rangle$ and final state $|f\rangle = |\phi_f\rangle \otimes \bigotimes_j |n'_j\rangle$. A transition via mode j requires that $\varepsilon_f = \varepsilon_i \pm \hbar\omega_j$ and $n'_j = n_j \mp 1$, with ε being the energy of the atomic states. That is, we either remove $\hbar\omega_j$ from the field, decreasing the photon number, and add it to the atom, which we call absorption, or we do the oppsite, which we call emission. Fermi's golden rule gives the transition probability rate for absorption and emission as

$$W_{i\to f} \approx \frac{2\pi}{\hbar}\left|\langle f|\,\hat{\mathbf{d}}\cdot\hat{\mathbf{E}}\,|i\rangle\right|^2 \delta(E_f^0 - E_i^0) \tag{4.6}$$

with

$$\langle f|\,\hat{\mathbf{d}}\cdot\hat{\mathbf{E}}\,|i\rangle = \langle\phi_f|\,\hat{\mathbf{d}}\,|\phi_i\rangle \cdot i\sum_j A_j\mathbf{e}_j \bigotimes_l \langle n'_l|\,(\hat{a}_j - \hat{a}^\dagger_j)\bigotimes_m |n_m\rangle\,. \tag{4.7}$$

For absorption via mode j we have $n'_j = n_j - 1$, and $i = g, f = e$, giving

$$W^j_{g\to e} \approx \frac{2\pi}{\hbar}\left|\langle e|\,\hat{\mathbf{d}}\cdot\mathbf{e}_j\,|g\rangle\right|^2 A_j^2 n_j \delta(\hbar\omega_j - \varepsilon_e + \varepsilon_g) \tag{4.8}$$

and for emission via mode j we have $n'_j = n_j + 1$ and $i = e, f = g$, giving

$$W^j_{e\to g} \approx \frac{2\pi}{\hbar}\left|\langle g|\,\hat{\mathbf{d}}\cdot\mathbf{e}_j\,|e\rangle\right|^2 A_j^2 (n_j+1) \delta(\hbar\omega_j - \varepsilon_e + \varepsilon_g). \tag{4.9}$$

Thus we can see that even if the field is in a vacuum state with $n_j = 0$, we can still observe spontaneous emission. This is in contrast to the classical field case, where only stimulated emission occurs. Spontaneous emission is thus a quantum mechanical effect, stemming from the interaction with vacuum fluctiations of the quantized electromagnetic field. The rate over all modes is given by summing over all modes, or equivalently integrating over the density of states, giving

$$W_{g\to e} = \frac{\omega_0\mathbf{d}^2}{3\pi\hbar c^3\epsilon_0}\bar{n}(\omega_0) \quad \text{and} \quad W_{e\to g} = \frac{\omega_0\mathbf{d}^2}{3\pi\hbar c^3\epsilon_0}[\bar{n}(\omega_0)+1], \tag{4.10}$$

where we have averaged over all polarization directions, and $\bar{n}(\omega_0)$ is the average photon number given by the Boltzmann distribution in Eq. (2.15). We can see that spontaneous emission dominates for low temperatures.

4.4 The Rabi Model

We will assume a coherent field with near resonance frequency between two atomic levels, and neglect al other modes. We can write the interaction Hamiltonian as $\hat{H}_I = \hat{V}_0 \cos\omega t$, and we define the Rabi frequency $\Omega_R = \hat{V}_0/\hbar$. Using the rotating wave approcimation, we can write the equations of motion for the probaiblity amplitudes in the interaction picture as

$$i\begin{pmatrix}\dot{C}_e\\ \dot{C}_g\end{pmatrix} = \frac{\Omega_R}{2}\begin{pmatrix}0 & \exp[i(\omega_0-\omega)t]\\ \exp[-i(\omega_0-\omega)t] & 0\end{pmatrix}\begin{pmatrix}C_e\\ C_g\end{pmatrix}. \tag{4.11}$$

Introducing the detuning $\Delta = \omega_0 - \omega$, the generalized Rabi frequancy $\Omega = \sqrt{\Delta^2 + \Omega_R^2}$ and solving the eigenvalue problem with initial condition $C_g(0) = 1$, $C_e(0) = 0$, we find the probability of being in the excited state as

$$P_e(t) = \frac{\Omega_R^2}{\Omega^2}\sin^2(\Omega t/2) = \frac{\Omega_R^2}{\Omega^2}\frac{1-\cos(\Omega t)}{2}, \tag{4.12}$$

which is called Rabi oscillations. The system oscillates between ground and a partially occupied excited state with the frequency Ω. If the detuning $\Delta = 0$, we observe full population inversion. Defining the pulse area $A = \Omega_R t$, we see that when $A = \pi$, calleda π-pulse, we have maximum population inverson. If $\Delta = 0$ with a pulse area of $A = \pi/2$, calleda $\pi/2$-pulse, we have a symmetric superposition of the ground and excited states.

4.5 Fully Quanum Mechanical Model: The Jaynes-Cummings Model

We consider a single mode quantized field interacting with a two-level system, where the electric field operator is

$$\mathbf{E} = \mathbf{e}\sqrt{\frac{\hbar\omega}{\varepsilon_0 V}}(\hat{a} + \hat{a}^\dagger)\sin kz \tag{4.13}$$

and defining $\hat{d} = \hat{\mathbf{d}} \cdot \mathbf{e}$ and $g = \sqrt{\hbar\omega/\varepsilon_0 V}\sin kz$ we write the interaction Hamiltonian as $\hat{H}_I = \hat{d}g(\hat{a} + \hat{a}^\dagger)$. We also introduce the atomic transition operators $\hat{\sigma}_+ = |e\rangle\langle g|$ and $\hat{\sigma}_- = |g\rangle\langle e|$ as well as the inversion operator $\hat{\sigma}_z = |e\rangle\langle e| - |g\rangle\langle g|$, obeying Pauli spin algebra. We rewrite the dipole operator as $\hat{d} = d(\hat{\sigma}_+ + \hat{\sigma}_-)$ and define the coupling $\lambda = dg/\hbar$. By defining $\Delta E = 0$, dropping the zero-point energy and using the rotating wave approximation, we can write the Jaynes-Cummings Hamiltonian as

$$\hat{H} = \frac{\hbar\omega_0}{2}\hat{\sigma}_z + \hbar\omega\hat{a}^\dagger\hat{a} + \hbar\lambda(\hat{\sigma}_+\hat{a} + \hat{\sigma}_-\hat{a}^\dagger). \tag{4.14}$$

The system will be in a product state with the atom in either the ground or excited state, and the field in a number state. The interaction will cause transition between the states $|g\rangle \otimes |n+1\rangle$ and $|e\rangle \otimes |n\rangle$, and thus $E_i = E_f$. Assuming $\Delta = 0$, we let $|i\rangle = |e\rangle \otimes |n\rangle$ and solve the time-dependent Schrödinger equation, we find the probabilities $P_i(t) = \cos^2(\lambda t\sqrt{n+1})$ and $P_f(t) = \sin^2(\lambda t\sqrt{n+1})$, and the atomic inversion is given by $\langle\hat{\sigma}_z\rangle = P_i(t) - P_f(t) = \cos\big(2\lambda t\sqrt{n+1}\big)$. We get a quantum Rabi frequency $\Omega_n = 2\lambda\sqrt{n+1}$. Interestingly, even when the field is in the vacuum state, oscillations still occur. The reasoning being that the atom emit and reabsorbs a photon. If we let the field be in a superposition of number states, as well as the atom in a superposition of the ground and excited state, we get in general, an entangled state. Considering coherent states, we find that the atomic inversion is given by $W(t) = e^{-\bar{n}}\sum_n \bar{n}^n/n!\cos(\Omega_n t)$, and the main oscillation is given by the Rabi frequency $\Omega_{\bar{n}}$. Since this Hamiltonian describes a two-level system in a single mode field, we should be able to describe a qubit in a microwave cavity. Is this model sufficient to describe this system for quantum technological applications? E.g. quantum computation or quantum sensing.

4.6 The Dressed States

Let $\Delta \neq 0$, and define the basis $\{|\psi_1^n\rangle = |e\rangle|n\rangle, |\psi_2^n\rangle = |g\rangle|n+1\rangle\}$. The Hamiltonian in this basis is given by

$$\hat{H}_n = \hbar\begin{bmatrix} n\omega + \omega_0/2 & \lambda\sqrt{n+1} \\ \lambda\sqrt{n+1} & (n+1)\omega - \omega_0/2 \end{bmatrix} \quad \text{with} \quad E_\pm^n = \hbar\omega(n+1/2) \pm \hbar\Omega_n(\Delta)/2, \tag{4.15}$$

where we define the Rabi frequency $\Omega_n(\Delta) = \sqrt{\Delta^2 + 4\lambda^2(n+1)}$. The eigenstates of this Hamiltonian are called the dressed states, and are given by

$$|n,+\rangle = \cos(\Phi_n/2)|e\rangle|n\rangle + \sin(\Phi_n/2)|g\rangle|n+1\rangle \tag{4.16}$$

$$|n,-\rangle = -\sin(\Phi_n/2)|e\rangle|n\rangle + \cos(\Phi_n/2)|g\rangle|n+1\rangle, \tag{4.17}$$

with $\Phi_n = \arctan(\Omega_n(0)/\Delta)$. Since the dressed states are stationary, we can use them to describe the time eveolution of a general state. Put an atom in state $|e\rangle$ into a general field, $|\psi(0)\rangle = \sum_n C_n|n\rangle|e\rangle$, writing this in terms of $|n,\pm\rangle$, and using the time evolution of the dressed states, we find

$$|\psi(t)\rangle = \sum_n C_n\left[\exp\big(-iE_+^n t/\hbar\big)\cos(\Phi_n/2)|n,+\rangle - \exp\big(-iE_-^n t/\hbar\big)\sin(\Phi_n/2)|n,-\rangle\right]. \tag{4.18}$$

4.7 The Density Operator Approach: Application to Thermal States

In general we might have a thermal field, that is mixed states, which requires a density matrix formulation. We assume that the system initially is not entangled so we may factor $\hat{\rho} = \hat{\rho}^A \otimes \hat{\rho}^F$, where $\hat{\rho}^A = |e\rangle\langle e|$ and $\hat{\rho}^F = \hat{\rho}_{Th}$. This gives the atomic inversion $W(t) = \sum_n P_n\cos(\Omega_n(0)t)$, with P_n described in Sec. 2.5. This atomic inversion collapses quickly from the excited states and then fluctates around zero. The fluctations are noise characteristic of thermal fluctuations, where for larger $\bar{n}$ the amplitude becomes smaller. Without assuming the thermal field, we can generally write the atomic inversion $W(t) = \sum_n 2\rho_{ee}^A(t) - 1$, with $\rho_{ee}^A(t) = \sum_n \langle n|\hat{\rho}^F(t)|n\rangle\cos^2(\Omega_n(0)t/2)$. It would be interesting to see how the atomic inversion would evolve for other types of fields or superposition of the intial state of the atom. We could maybe also imagine two two-level systems inside the cavity that are entangled, which would be interesting to study.

5 Quantum Coherence Functions

5.1 Classical Coherence Functions

Imagine a two slit experiment. For a source with bandwidth $\Delta\omega$ and path difference Δs, coherence will occur for $\Delta s \leq c/\Delta\omega$, with $\Delta s_{\text{coh}} = c/\Delta\omega$ the coherence length, and $\Delta t_{\text{coh}} = \Delta s_{\text{coh}}/c$ the coherence time. Let K_1, K_2 be complex geometric factors corresponding to the path lengths. The ergodic hypothesis tell us that the time average is equivalent to the ensemble average. Using $x_i = \mathbf{r}_i, t_i$ we can write the first order coherence function

$$\gamma^{(1)}(x_1, x_2) = \frac{\langle E^*(x_1)E(x_2)\rangle}{\sqrt{\langle |E(x_1)|^2\rangle\langle |E(x_2)|^2\rangle}}, \tag{5.1}$$

and the intensity at the slits is defined as $I_i = |K_i|^2\langle |E(x_i)|^2\rangle$. Writing $K_j = |K_j|\exp(i\psi_j)$ and $\gamma^{(1)}(x_1, x_2) = \left|\gamma^{(1)}(x_1, x_2)\right|\exp(i\Phi_{12})$ we have the intensity at the detector

$$I(\mathbf{r}) = I_1 + I_2 + 2\sqrt{I_1 I_2}\left|\gamma^{(1)}(x_1, x_2)\right|\cos(\Phi_{12} - \psi) \tag{5.2}$$

, with $\psi = \psi_1 - \psi_2 < \Delta s_{\text{coh}}$. Furthermore, $\left|\gamma^{(1)}\right| = 1$ is complete coherence, $0 < \left|\gamma^{(1)}\right| < 1$ is partial coherence, and $\left|\gamma^{(1)}\right| = 0$ is complete decoherence. Reyleigh's fringe visibility is then $\mathcal{V} = 2\sqrt{I_1 I_2}\left|\gamma^{(1)}\right|/(I_1 + I_2)$. For a source with coherence time τ_0, inversly related to the spectral line width, and assuming a Lorentzian power spectrum, we obtain the autocoherence function $\gamma^{(1)}(\tau) = \exp(i\omega_0\tau - |\tau|/\tau_0)$. Why do we so often see a Lorentzian distribution in atomic physics? What is the physical explanation of it?

5.2 Quantum Coherence Functions

We consider a detector constructed from a single atom smaller than the wavelength (dipole approximation). Absorption leads to ionization and detection of the photoelectron, thus detecting the photon. The detector is coupled to the field by the dipole interaction (Sec 4.3), and we assume that the excited photoelectron may be described by the wave function of a plane wave. We can define the first order correlation function as $G^{(1)}(x_1, x_2) = \text{tr}\left(\hat{\rho}\hat{E}^-(x_1)\hat{E}^+(x_2)\right)$, where $\hat{\rho}$ is the density function of the field and $\hat{E}^+$ is the positive part of the field as defined in Sec. 2.4. The detector response from light arriving from slit i is $G^{(1)}(x_i, x_i) = \text{tr}\left(\hat{\rho}\hat{E}^-(x_i)\hat{E}^+(x_i)\right)$. The normalized quantum coherence function is $g^{(1)}(x_1, x_2) = G^{(1)}(x_1, x_2)/\sqrt{G^{(1)}(x_1, x_1)G^{(1)}(x_2, x_2)}$, with the same types of coherences as in the classical case. For a single mode number state, or coherent state we get that $\left|g^{(1)}\right| = 1$.

5.3 Young's Interference

If we have a laser beam incident on two pinholes close together, we will observe an interference pattern with fringes for angles $\Phi = 2\pi m$ with m an integer. The intensity falls of at a speed proportional to $1/r^2$ from the central fringe. If the incoming light is a number state or a coherent state, the interference pattern is the same, only the max intensity being shifted by some amount due to the average photon number. Note that the result will be different if the light incident on each pinhole comes from different sources.

5.4 Higher-Order Coherence Functions

First order coherence experiments cannot distinguish between states with different photon number distributions if their spectral distributions are the same. Hanbury Brown and Twiss developed a coherence experiment relying on the correlation of intensities instead of the fields as Young had done. The experiment uses a beam splitter and a time delay for one of the beams, and if the time delay, τ, is shorter than the beam's coherence time, it is possible to determine the photon statistics of the beam. We find that the second order coherence function is $\gamma^{(2)}(\tau) = \langle I(t)I(t+\tau)\rangle/\langle I(t)\rangle^2$ if the fields are stationary, and more generally $\gamma^{(2)}(x_1, x_2; x_2, x_1) = \langle I(x_1)I(x_2)\rangle/\langle I(x_1)\rangle\langle I(x_2)\rangle$. For chaotic light we find that $1 \leq |\gamma^{(2)}(\tau)| \leq 2$ and the Lorentzian spectrum $\gamma^{(2)}(\tau) = 1 + \exp(-2|\tau|/\tau_0)$. When the time delay goes to zero we get twice the detection rate compared to long times, we say that the photons arrive in bunched pairs, called the photon bunching effect, or the Hanbury Brown and Twiss effect. Introducing the second order quantum correlation function $G^{(2)}(x_1, x_2; x_2, x_1) = \text{tr}\left(\hat{\rho}\hat{E}^-(x_1)\hat{E}^-(x_2)\hat{E}^+(x_2)\hat{E}^+(x_1)\right)$ interpreted

as the ensemble average $I(x_1)I(x_2)$ and is normal ordered, we can write the second order quantum coherence function as $g^{(2)}(x_1, x_2; x_2, x_1) = G^{(2)}(x_1, x_2; x_2, x_1)/G^{(1)}(x_1, x_1)G^{(1)}(x_2, x_2)$. It is second order coherent if $|g^{(1)}| = 1$ and $g^{(2)} = 1$. For a single mode field we have $g^{(2)}(\tau) = g^{(2)}(0) = 1$ which implies the same relation for a coherent state. For a single mode thermal field we have $g^{(2)}(\tau) = g^{(2)}(0) = 2$, and for a multimode thermal field $g^{(2)}(\tau) = 1 + |g^{(1)}(\tau)|$. Collision-broadened light gives the Lorentzian spectrum $g^{(2)}(\tau) = 1 + \exp(-2|\tau|/\tau_0)$ just as in the classical case. A single mode field in a number state can exhibit $g^{(2)}(0) < 1$, which is classically forbidden, and is a sub-Poissonian state. We can characterize states with Mandel's Q-parameter

$$Q = \frac{\langle(\Delta\hat{n})^2\rangle - \langle\hat{n}\rangle}{\langle\hat{n}\rangle}, \tag{5.3}$$

for which states with $Q = 0$ are Poissonian, $Q > 0$ are super-Poissonian, and $-1 < Q < 0$ are sub-Poissonian. What properties does these types of states share? If two states are constructed in different ways, but both are sub-Poissonian for example, what properties do they share?

6 Beam Splitters and Interferometers

6.1 Experiments with Single Photons

To explain the photoelectric effect with all its aspects the field must be treated quantum mechanically. Grangier created a source able to produce anti-correlated photons, with the help of a beam splitter. In contrast to Taylor, who's source of photon was thermal and thus exhibited bunching effects, leading to misinterpretations.

6.2 Quantum Mechanics of Beam Splitters

For a classical light field with amplitude ε_1 incident on a lossless beam splitter, the resulting reflected field is $\varepsilon_2 = r\varepsilon_1$ and the resulting transmitted field is $\varepsilon_3 = t\varepsilon_1$ with r, t complex. Since the beam splitter is lossless we have $|\varepsilon_1|^2 = |\varepsilon_2|^2 + |\varepsilon_3|^2$ or equivalently $|r|^2 + |t|^2 = 1$. Treating the beam splitter quantum mechanically instead we recall that we have an incident vacuum state on the currently unused plane of the beam splitter. Replacing ε_i with $\hat{a}$ gives $\hat{a}_2 = t'\hat{a}_0 + r\hat{a}_1$ and $\hat{a}_3 = r'\hat{a}_0 + t\hat{a}_1$, we can write this as a matrix equation

$$\begin{pmatrix}\hat{a}_2\\ \hat{a}_3\end{pmatrix} = \mathbf{B}\begin{pmatrix}\hat{a}_0\\ \hat{a}_1\end{pmatrix} \quad \text{with} \quad \mathbf{B} = \begin{pmatrix}t' & r\\ r' & t\end{pmatrix}, \tag{6.1}$$

where $\mathbf{B}$ is the scattering matrix and r, r', t, t' follow the reciprocity relations: $|r'| = |r|, |t'| = |t|, |r|^2 + |t|^2 = 1, r^*t' + r't^* = 0$ and $r^*t + r't'^* = 0$. These relations are needed to preserve the commutation relations of the beam splitter. We also define the transmissivity $T = |t|^2$ and reflectivity $R = |r|^2$. The photon number, together with energy, is conserved through the beam splitter. For $0 \leq \vartheta \leq \pi$ and $0 \leq \varphi \leq 2\pi$ we can rewrite the scattering matrix using $t = t' = \cos(\vartheta/2)$ and $r = -r'^* = \exp(i\varphi)\sin(\vartheta/2)$, where ϑ determines the transmissivity and reflectivity, while φ determines the phase shift introduced on the reflected beam, which we take to be $\varphi = \pi/2$. How easy is it to manufacture a mirror with a known phase-shift like this? For a number state incident we get an entangled output state, while if the incident state is a coherent state we do not get an entangled state. Note that the limiting case of a single incident photon is not possible to arrive to from the coherent state. For a 50:50 beam splitter the with two incident photons, one on each face, the output will be either $i(|2\rangle|0\rangle + |0\rangle|2\rangle)/\sqrt{2}$, that is, we will never detect one photon in each detector, but only two photons in either. This comes from photon interference, and was demonstrated by Hong, Ou, and Mandel. This effect is not general if we have n incident photons on each face of the beam splitter, however the output state with $2n$ still are the most likely. The resulting states of n incident photons are called arcsine states. The interference of one incident photon together with an incident coherent state will dramatically change the outgoing photon distribution compared to a vacuum state. It would be interesting to find a physical explanation for why a macroscopic amount of photons ($|\alpha|^2 = \bar{n}$ large) is so affected by just a single photon.

6.3 Interferometry with a Single Photon

In a Mach-Zehnder interferometer we have a beam splitter and both transmission and reflection has a mirror leading to another beam splitter with two detectors. The detectors cannot distinguish which way the photon took in the first beam splitter. One of the ways has a phase shifter. Sending in a single photon

(having the vacuum state incident on the second face), gives the probability of detecting a photon in the first detector as $P_1 = (1 - \cos\vartheta)/2$ and the second detector $P_2 = (1 + \cos\vartheta)/2$, where ϑ is the relative phase between the two ways, imparted by the phase shifter. That is, the single photon interferes with itself.

6.5 Interferometry with Coherent States of Light

We use the same Mach-Zehnder interferometer as for a single photon but switch the single-photon to a coherent state, and define the number difference operator $\hat{O} = \hat{a}^\dagger\hat{a} - \hat{b}^\dagger\hat{b}$, where $\hat{a}^\dagger\hat{a}$ corresponds to the number of detected photons in detector one and $\hat{b}^\dagger\hat{b}$ the number of detected photons in detector 2. The expectation value for the number difference operator with respect to the resulting state after the two beam splitters is $\langle\hat{O}\rangle = |\alpha|^2\cos\vartheta$, which is similar to coherent classical light. The phase uncertainty, or the sensitivity of the measurement, is defined as $\Delta\vartheta = \Delta\hat{O}/\partial_\varphi\langle\hat{O}\rangle = 1/\sqrt{\bar{n}}|\sin\varphi|$, with φ being the relative phase. These types of states, especially squeezed states (which will be discussed in Chapter 7) is of particular interest due to the resulting entanglement.

6.6 The SU(2) Formulation of Beam Splitters and Interferometry

Beam splitters and interferometers may be described in the language of quantum angular momentum algebra. Defining $\hat{J}_+ = \hat{a}_0^\dagger\hat{a}_1$, $\hat{J}_- = \hat{J}_+^\dagger$ and $\hat{J}_3 = (\hat{a}_0^\dagger\hat{a}_0 - \hat{a}_1^\dagger\hat{a}_1)/2$, we get $[\hat{J}_+, \hat{J}_-] = 2\hat{J}_3$, $\hat{J}_1 = (\hat{J}_+ + \hat{J}_-)/2$, and $\hat{J}_2 = (\hat{J}_+ - \hat{J}_-)/2i$ with $[\hat{J}_i, \hat{J}_j] = i\varepsilon_{ijk}\hat{J}_k$ for $i, j, k \in \{1, 2, 3\}$. This is the familiar angular momentum algebra in quantum mechanics, also known as special unitary group of degree 2 in Lie algebra, SU(2). This allows us to describe the unitary evolution of a beam splitter as the rotation about an axis. Generally we can write $\hat{U}(\vartheta, \varphi) = \exp\Big[(\hat{J}_+ e^{i\varphi} - \hat{J}_- e^{-i\varphi})\vartheta/2\Big]$, and as an example, for a 50/50 beam splitter ($\vartheta = \pi/2$) with the mirror imparting a $\pi/2$ phase shift ($\varphi = \pi/2$) we have $\hat{U}(\pi/2, \pi/2) = \exp\Big(i\pi\hat{J}_1/2\Big)$. Thus, the beam splitter is a $\vartheta = \pi/2$ rotation about the $\hat{J}_1$ axis. With this language we can describe interferometers, which gives the same result as earlier in the chapter.

6.8 Photons Do Not Interfere

What interferes are quantum amplitudes, not the photons, which are described by wave packets. We note that Dirac's phrase: "Each photon interferes only with itself. Interference between two photons never occurs." Should not be taken as truth. It is instead the quantum amplitudes associated with the quantized field and hence the paths in the interferometer that causes single photon interference, and multiphoton interference means interference with states of light containing more than one photon.

6.9 Are Photons Entangled

Single photons themselves are not entangled, however, the field modes of the single photon can be entangled with each other. Say a single photon passes through a beam splitter, the resulting field modes (reflected or transmitted) will be entangled and spatially separated.

7 Nonclassical Light

All light states are nonclassical, but it can be hard to see the quantum mechanical effects. The only pure light state that have a $P(\alpha)$ distribution that is positive everywhere and note more singular than a delta function is the coherent state.

7.1 Quadrature Squeezing

Two operators $\hat{A}, \hat{B}$ with commutation relation $[\hat{A}, \hat{B}] = i\hat{C}$ will have $\langle(\Delta\hat{A})^2\rangle\langle(\Delta\hat{B})^2\rangle \geq |\langle\hat{C}\rangle|^2/4$. We call the system squeezed if $\langle(\Delta\hat{A})^2\rangle < |\langle\hat{C}\rangle|/2$ (or similar for $\hat{B}$). Ideal squeezed states are states for which $[\hat{X}_1, \hat{X}_2] = i/2$ holds. Quadrature squeezing exist when $\hat{A} = \hat{X}_1, \hat{B} = \hat{X}_2, \hat{C} = 1/2$ and $\langle(\Delta\hat{X}_1)^2\rangle < 1/4$ (or similar for $\hat{X}_2$). For this to hold we require that the P-function is nonpositive somewhere. We can introduce a general quadrature operator $\hat{X}(\vartheta) = [\hat{a}\exp(i\vartheta) + \hat{a}^\dagger\exp(-i\vartheta)]/2$ from where $\hat{X}_1$ and $\hat{X}_2$ follow. For the parameter $s(\vartheta) = 4\langle(\Delta\hat{X}(\vartheta))^2\rangle - 1$, we have squeezed states when $-1 \leq s(\vartheta) < 0$.

We can generate squeezed states by introducing a squeezing operator $\hat{S}(\xi) = \exp\big[(\xi^*\hat{a}^2 - \xi(\hat{a}^\dagger)^2)/2\big]$, where $\xi = r\exp(i\vartheta)$, with r the squeeze parameter, and $0 \leq r < \infty, 0 \leq \vartheta \leq 2\pi$. Where does this operator come from? The squeezing operator can be interpreted as a two-photon displacement operator. We clearly have $\hat{S}^\dagger(\xi) = \hat{S}(-\xi)$ and the Baker-Housdorff lemma gives $\hat{S}^\dagger(\xi)\hat{a}\hat{S}(\xi) = \hat{a}\cosh r - \hat{a}^\dagger\exp(i\vartheta)\sinh r$ and $\hat{S}^\dagger(\xi)\hat{a}^\dagger\hat{S}(\xi) = \hat{a}^\dagger\cosh r - \hat{a}\exp(-i\vartheta)\sinh r$. Applying the squeezing operator to the vacuum state we get $|\xi\rangle = \hat{S}(\xi)|0\rangle$, and calculating the fluctuations of the quadrature operator for $\vartheta = 0$ yields $\langle(\Delta\hat{X}_1)^2\rangle = \exp(-2r)/4$ and $\langle(\Delta\hat{X}_2)^2\rangle = \exp(2r)/4$. Thus, we have squeezing in the $\hat{X}_1$ quadrature, choosing $\vartheta = \pi$ we get squeezing in the $\hat{X}_2$ quadrature, for an arbitrary ϑ we can rotate the coordinate system and find a similar result. Applying also the displacement operator $|\alpha,\xi\rangle = \hat{D}(\alpha)\hat{S}(\xi)|0\rangle$ we obtain a squeezed state that is displaced from the origin by α.

The electric field of a squeezed state will have phase-dependant noise. Similar to how we generated an electric field by rotating the coherent state around the origin, with the uncertainty in the field coming from the uncertainty in the coherent state's quadratures. Since a squeezed state has less uncertainty in one of the quadratures and more in the other, the projection while rotating around the origin will have very different uncertainties. For example, this can be used to obtain an electric field where the peak position has very little uncertainty. This fact can be used in the detection of very weak signals, like gravity wave detector. It would be interesting to use these state in quantum sensing applications.

A squeezed state on the form $|\alpha,\xi\rangle$ can be characterized by the correlation between the quadratures. Also, important to note is that the squeezed state $|\alpha,\xi\rangle$ is not every possible squeezed state. What other types of squeezed states are there?

7.2 Generation of Quadrature Squeezed Light

To generate quadrature squeezed light most schemes use degenerate parametric down-conversion, realized by nonlinear optics. If a certain nonlinear medium is pumped by a laser with frequency ω_p it produces two photons with frequency $\omega/\omega_p/2$ with the Hamiltonian $\hat{H} = \hbar\omega\hat{a}^\dagger\hat{a} + \hbar\omega_p\hat{b}^\dagger\hat{b} + i\hbar\chi^{(2)}(\hat{a}^2\hat{b}^\dagger - (\hat{a}^\dagger)^2\hat{b})$, with a the signal mode, b the pump mode, and $\chi^{(2)}$ is the second order susceptibility. The parametric approximation says that the pump field is in a strong classical coherent field that remains undepleted of photons during the relevant timescale. Assuming the coherent state and approximation $\hat{b} = \beta\exp(-i\omega_p t)$ we can write the Hamiltonian in the interaction picture $\hat{H}_I(t) = i\hbar\{\eta^*\hat{a}^2\exp[i(\omega_p - 2\omega)t] - \eta(\hat{a}^\dagger)^2\exp[-i(\omega_p - 2\omega)t]\}$ with $\eta = \beta\chi^{(2)}$. The time dependency may be removed by choosing a material such that $\omega_p = 2\omega$ giving the Hamiltonian $\hat{H}_I = i\hbar(\eta^*\hat{a}^2 - \eta(\hat{a}^\dagger)^2)$, with the unitary evolution operator $\hat{U}(t,0) = \exp\big(\eta^* t\hat{a}^2 - \eta t(\hat{a}^\dagger)^2\big)$ which is the squeezing operator for $\xi = 2\eta t$. There is a similar process called degenerate four-wave mixing that may also be used.

7.3 Detection of Quadrature Squeezed Light

To detect or measure squeezed states we can use a method called balanced homodyne detection. We use a 50/50 beam splitter with the $\hat{a}$ input state being the signal we want to measure, and $\hat{b}$ a strong coherent laser field with the same frequency as the signal (homodyne) but with adjustable phase. The two output $(\hat{c},\hat{d})$ intensities of the beam splitter is measured, and a correlator takes the difference. Assuming the coherent state $\left|\beta e^{-i\omega t}\right\rangle$ and $\beta = |\beta|\exp(i\psi)$ with ψ the phase, and defining $\vartheta = \psi + \pi/2$, we get that the number difference in the detectors is $\langle n_{cd}\rangle = \langle\hat{c}^\dagger\hat{c} - \hat{d}^\dagger\hat{d}\rangle = 2|\beta|\langle\hat{X}(\vartheta)\rangle$. With this we can measure and arbitrary quadrature by tuning the phase of the laser (changing ϑ).

7.4 Amplitude (or Number) Squeezed States

The heuristic number phase commutator $[\hat{n},\hat{\varphi}] = i$ lead to the uncertainty relation $\Delta n\Delta\varphi \geq 1/2$. Thus, by analogy to the quadrature squeezing we can imagine states squeezed in number such that $\Delta n < \sqrt{\bar{n}}$ or in phase such that $\Delta\varphi < 1/(2\sqrt{\bar{n}})$. We note that there does not exist a Hermitian phase operator, which create problems. We can still define some photon statistics for amplitude squeezed states, which are nonclassical and sub-Possonian. I don't like using the phase like this since we lack a phase-operator. It feels weird, and I do not understand why we can do this. A way to categorize states is with Mandel's Q-parameter $Q = (\langle(\Delta\hat{n})^2 - \langle\hat{n}\rangle\rangle)/\langle\hat{n}\rangle$. For $-1 \leq Q < 0$ the states are said to be sub-Poissonian and for $Q = 0$ the states are Poissonian, and for $Q > 0$ super-Poissonian. A coherent state have $Q = 0$ and thus sub-Poissonian states have a narrower distribution and super-Poissonian states have a broader distribution than a coherent state. It is also possible for a state to be both number squeezed and quadrature squeezed.

7.5 Photon Antibunching

Antibunching is when the probability for joint detection increases with time which is a nonclassical effect, that is $g^{(2)}(\tau) > g^{(2)}(0)$. The second order coherence function is related to the Q-parameter for a single mode field by $Q = \langle \hat{n} \rangle [g^{(2)}(0) - 1]$ and since $g^{(2)}(\tau) = g^{(2)}(0) =$ constant we can not see bunching or antibunching effects for a single mode field, since we lack the time dependence. To see this we need multimode fields which will give the time dependence. For a driving field with Rabi frequency Ω and spontaneous emission decay rate γ, for $\Omega \gg \gamma$ we get $g^{(2)}(\tau) = 1 - \exp(-3\gamma\tau/4)\cos(\Omega t)$ and for $\Omega \ll \gamma$ we get $g^{(2)}(\tau) = [1 - \exp(-\gamma\tau/2)]^2$. Both of which have $g^{(2)}(0) = 0$ and $g^{(2)}(\tau) > 0$ for all $\tau > 0$, and thus showing antibunching.

7.6 Schrödinger Cat-States

A Schrödinger cat state is a state on the form $|Psi\rangle = N(|\alpha\rangle + \exp(i\Phi)\,|-\alpha\rangle)$ where N is a normalization constant, with large $|\alpha|$ the states are macroscopically distinguishable. The question is why we cannot see macroscopic entanglement in everyday life (some recent experiments have managed to create smaller macroscopic states of this form)? No quantum state is truly isolated, especially not a macroscopic one, and thus, if a pure macroscopic state is created it would quickly interact with the environment and decohere into a classical statistical mixture, or a mixed state. Depending on the chosen phase there are three main types of cat states. For $\Phi = 0$ we have the even states $|\Psi_e\rangle = N_e(|\alpha\rangle + |-\alpha\rangle)$, for $\Phi = \pi$ we have the odd states $|\Psi_o\rangle = N_o(|\alpha\rangle - |-\alpha\rangle)$, and finally for $\Psi = \pi/2$ we have the Yurke-Stoler states $|\Psi_{YS}\rangle = (|\alpha\rangle + i\,|-\alpha\rangle)/\sqrt{2}$. All of these states are eigenstates to $\hat{a}^2$ with eigenvalue α^2.

The density matrix for the statistical mixture is defined as $\hat{\rho}_{\text{mixture}} = (|\alpha\rangle\langle\alpha| + |-\alpha\rangle\langle-\alpha|)/2$, and the (for example) even cat state is $\hat{\rho}_e = |N_e|^2(|\alpha\rangle\langle\alpha| + |\alpha\rangle\langle-\alpha| + |-\alpha\rangle\langle\alpha| + |-\alpha\rangle\langle-\alpha|)$. To distinguish between these two states we must be able to detect the coherences. This may be done using photon statistics as for the even cat state $P_n = 0$ for n odd while P_n is nonzero for n even, while for the mixed state it is nonzero for all n, a similar argument can be done for the odd cat state. The Yurke-Stoler state is also nonzero for all n. Both the even cat state and the Yurke-Stoler cat state exhibit quadrature squeezing in $\hat{X}_2$, while there is no squeezing for the odd cat state and the mixed state. How are cat states used as qubits? I've heard that they can be.

The Lindblad (CKSL) master equation is $\partial_t\hat{\rho} = -i/\hbar \times [\hat{H}, \hat{\rho}(t)] + \sum_\alpha \gamma_\alpha(\hat{L}_\alpha\hat{\rho}(t)\hat{L}_\alpha^\dagger - \{\hat{L}^\dagger\hat{L}, \hat{\rho}(t)\}/2)$, where $\hat{L}$ is jump operator and γ a parameter related to the probability (or rate) of the jump occurring and can either be calculated with Fermi's golden rule or fit to experimental data. The jump operator is an operator that transfer one quantum state into another. Examples can be the loss or gain of a photon in a cavity or the excitation or deexcitation of an atom.

Appendix D: Nonlinear Optics and Spontaneous Parametric Down-Conversion

The physical effect of down-conversion is due to the fact that in nonlinear optics the induced polarization is strong enough to need at least second order susceptibility to be considered. That is the linear response in polarization is not sufficient for these materials. Noting here that many materials used for down-conversion are anisotropic. Does there exist materials that can be used for down-conversion that are isotropic? The parametric approximation tells us that we pump with a very strong coherent laser field that can be treated classically as long as the interactions are short enough that the pump photons don't run out. Doing this we get the phase matching conditions $\omega_p = \omega_s + \omega_i$ and $\mathbf{k}_p = \mathbf{k}_s + \mathbf{k}_i$, where p is for pump, s for signal, and i for idler.

9 Optical Test of Quantum Mechanics

Today we have the capability to perform experiments on single atoms, molecules, or ions, contradicting what Schrödinger thought would be possible. Fundamental tests are tests that distinguish between quantum mechanical theory and hidden-variable theories.

9.1 Photon Sources: Spontaneous Parametric Down-Conversion

We consider the non-degenerate case for parametric down-conversion with the interaction Hamiltonian $\hat{H}_I \sim \chi^{(2)}\hat{a}_p\hat{a}_s^\dagger\hat{a}_i^\dagger$ + h.c. with p, s, i being the pump, signal and idler. Imagining the simplest case $|1\rangle_p\,|0\rangle_s\,|0\rangle_i \to |0\rangle_p\,|1\rangle_s\,|1\rangle_i$, where the two photons are created almost instantaneously in the signal

and idler, and which is called spontaneous since the signal and idler originally are in the vacuum state. We must respect conservation and thus, $\omega_p = \omega_s + \omega_i$ and inside the crystal we must have $\mathbf{k}_p \approx \mathbf{k}_s + \mathbf{k}_i$, where the approximation stems from the uncertainty in crystal length. These two conditions are called the phase-matching conditions and only non-centrosymmetric crystals have a non-vanishing $\chi^{(2)}$.

There are two types of spontaneous parametric down-conversion. In type I the signal and idler have the same polarization and which is orthogonal to the pump. The outgoing light will be two concentric cones, one for the signal and one for the idler. A single down-conversion will come out on opposite sides of the two cones. That is, if the signal comes out at $\pi/2$ (measured from the x-axis) the idler comes out at $3\pi/2$. In type II the signal and idler are orthogonal, which creates two off-centre cones, the ordinary and extraordinary ray, with an intersection in two points. Photons emerging at these two points will be polarization entangled. I don't really understand why neither cone is not concentric on the pump axis. It should have something to do with the phase-matching I think? Type II down-conversion may be used to create the bell states $|\Psi^{\pm}\rangle = (|H\rangle_1 |V\rangle_2 \pm |V\rangle_1 |H\rangle_2)/\sqrt{2}$ and $|\Phi^{\pm}\rangle = (|H\rangle_1 |H\rangle_2 \pm |V\rangle_1 |V\rangle_2)/\sqrt{2}$.

9.2 The Hong-Ou-Mandel Interferometer

Two single-photon states incident on a beam splitter will experience bunching effects, and detectors will never measure coincidences. An interpretation of this is that the photons are incident on the beam splitter simultaneously. I do not understand this interpretation. Hong-Ou-Mandel tested this effect by using type I down-conversion. By creating two photons using down-conversion and then aiming them on mirrors to a beam splitter and moving the beam splitter slightly one could measure the time-difference of arrival. Hong-Ou-Mandel (HOM) did this experiment with great success and manages to confirm theory. They did not reach 0 coincidences with the beam splitter at equal distances, which is contributed to an alignment error of the beams.

9.3 The Quantum Eraser

The photons in the HOM experiment are being created by the same source of the type I down-conversion and thus have the same polarization, and being indistinguishable is the reason for the interference. Put another way, the inability to detect which path the photon has taken causes interference. Placing a rotator at the idler beam and rotating it to vertical polarization will completely destroy the interference, causing classical statistics to dictate the count rate ($P_{\text{coincidence}} = 0.5$). Even if the polarization is not measured in the detector, the possibility of detecting polarization, and thus which path the photon took, removes the interference.

Introducing linear polarizers at the detectors with angles ϑ_1, ϑ_2, we get the probability to detect coincidences $P_{\text{coincidence}} = \sin^2(\vartheta_2 - \vartheta_1)/4$. That is, we can choose the angles such that $P_{\text{coincidence}} = 0$, regaining the bunching effect and the interference. We have erased the path-information. This work by the polarizers projecting the state onto two orthogonal states, thus removing the possibility of double counts. We note that the process of erasing the path-information take place outside the HOM-interferometer and after the photons have crossed paths. That is, the photons do not interact in any way again after the polarizers.

9.5 Superluminal Tunnelling of Photons

Imagining once again a balanced HOM-interferometer, but now with a dielectric mirror, acting as a tunnelling barrier at the signal beam. Interestingly, by the insertion of this barrier to regain a balance in coincidence, we must lengthen the signal path, not the idler path. The important distinction being that the tunnelling through the mirror happens superluminally. The common explanation for this phenomenon is that the transmitted Gaussian wave packet is created by pulse reshaping of the front of the Gaussian wave packet before interacting with the barrier. That is, the front of the pulse is being reshaped to new Gaussian pulse and transmitted before the peak of the original pulse have hit the barrier. This can not be used for faster-than-light signalling. Is this the same as fast light that can be seen in materials with negative refractive indices. For example Rare-Earth ion-crystals.

9.6 Optical Test of Local Realistic Theories and Bell's Theorem

It is possible to produce any of the four Bell states using either type I or type II down-conversion. For a birefringent crystal, the ordinary beam is along the path of the input beam while the extraordinary beam changes direction. The refraction is different for different polarizations. For photons passing through the

birefringent crystal with an angle ϑ the o-ray will be in the state $|\vartheta\rangle$ while the e-ray will be in the state $|\vartheta_\perp\rangle$.

For two crystals with angle ϑ and φ that are placed in front of the ordinary or extraordinary beams, we can define the Bell state

$$\left|\Psi^-\right\rangle = \frac{1}{\sqrt{2}}[(\cos\vartheta\sin\varphi - \sin\vartheta\cos\varphi)\left|\vartheta\right\rangle_1\left|\varphi\right\rangle_2 + (\cos\vartheta\cos\varphi + \sin\vartheta\sin\varphi)\left|\vartheta\right\rangle_1\left|\varphi_\perp\right\rangle_2 \tag{9.1}$$

$$-(\sin\vartheta\sin\varphi + \cos\vartheta\cos\varphi)\left|\vartheta_\perp\right\rangle_1\left|\varphi\right\rangle_2 - (\sin\vartheta\cos\varphi - \cos\vartheta\sin\varphi)\left|\vartheta_\perp\right\rangle_1\left|\varphi_\perp\right\rangle_2]. \tag{9.2}$$

For $\vartheta = \varphi$ the Bell state $|\Psi^-\rangle = (|\vartheta\rangle_1|\vartheta_\perp\rangle_2 - |\vartheta_\perp\rangle_1|\vartheta\rangle_2)/\sqrt{2}$. We can test quantum mechanics against local hidden variable theories assuming space-light, space-time separation. Local hidden variable theories essentially say that we lack some information about the system (a hidden variable) that determines if the state is $|\vartheta\rangle_1|\vartheta_\perp\rangle_2$ or $|\vartheta_\perp\rangle_1|\vartheta\rangle_2$, and the measurement merely shows the outcome of this hidden variable. Another interpretation is that the act of measuring on of the photons forces the collapse of the wave function, and that this information somehow is transferred to the other photon, seemingly violating the speed of light, and is thus non-local.

We define two operators $\hat{A}(\vartheta)$ and $\hat{B}(\varphi)$, with value $+1$ if the photon emerges in the ordinary beam and -1 if the photon emerges in the extraordinary beam for the two birefringent crystals, where ϑ, φ is the angles for the different crystals. We then define the correlation function $C(\vartheta,\varphi) = \langle\hat{A}(\vartheta)\hat{B}(\varphi)\rangle$, which for the state $|\Psi^-\rangle$ is $C(\vartheta,\varphi) = -\cos[2(\vartheta-\varphi)]$, which is a quantum mechanical result. For a local hidden-variable theory we introduce the hidden variable λ which affects the outcome of the measurement, and thus we have $\hat{A}(\vartheta,\lambda)$ and $\hat{B}(\varphi,\lambda)$, with the values ±1 still. If $\hat{\rho}(\lambda)$ is the probability distribution of the hidden variable we obtain the correlation function in local hidden variable theory as $C_{\mathrm{HV}}(\vartheta,\varphi) = \int d\lambda\hat{\rho}(\lambda)\hat{A}(\vartheta,\lambda)\hat{B}(\varphi,\lambda)$. Introducing the quantity

$$S = \int d\lambda\hat{\rho}(\lambda)(\hat{A}(\vartheta,\lambda)[\hat{B}(\varphi,\lambda) + \hat{B}(\varphi',\lambda)] + \hat{A}(\vartheta',\lambda)[\hat{B}(\varphi,\lambda) - \hat{B}(\varphi',\lambda)]) = \pm2 \tag{9.3}$$

we have that

$$-2 \leq C_{\mathrm{HV}}(\vartheta,\varphi) + C_{\mathrm{HV}}(\vartheta',\varphi) + C_{\mathrm{HV}}(\vartheta,\varphi') - C_{\mathrm{HV}}(\vartheta',\varphi') \leq 2, \tag{9.4}$$

which is the Clauser-Horne-Shimony-Holy-inequality (CHSH-inequality). However, choosing the angles $\vartheta = 0, \vartheta' = \pi/4, \varphi = \pi/8$, and $\varphi' = -\pi/8$ gives the quantum mechanical $S_{\mathrm{QM}} = C(\vartheta,\varphi) + C(\vartheta',\varphi) + C(\vartheta,\varphi') - C(\vartheta',\varphi') = 2\sqrt{2}$, that is, quantum mechanics have results that no local hidden variable theory can explain.

9.7 Franson's Experiment

Imagine two photons, signal and idler, simultaneously being created by down-conversion and aimed in two directions. There is no single photon interference since the photon coherence times are much shorter than the difference in path lengths in the arms of the interferometer. The path difference in the arms comes from a beam splitter leading the photon to an adjustable length $L_{s/i}$ or a shorter length S. The paths (L_s, L_i) and (S, S) are indistinguishable since the detectors click simultaneously, and we do not know when the photons were created, while (L_s, S) and (S, L_i) are distinguishable and rejected to obtain the reduced state $|\psi\rangle = (|S\rangle_s|S\rangle_i + e\exp(i\Phi)|L_s\rangle_s|L_i\rangle_i)/2$. The angle Φ is the relative phase between the two path processes, $\Phi \approx \omega_p(\Delta L_s + \Delta L_i)/2c$. We find that the probability of coincidence detection is $P_{\mathrm{coincidence}} = [1 + \cos(\omega_p(\Delta L_s + \Delta L_i)/2c)]/2$. If the correlation fringes have a greater visibility than 70.7 % it violates Bell's inequality. Experiments have measures a visibility of 80.4(6) %. I have a hard time completely understanding this experiment. But the results seem significant.

9.8 Application of Down-Converted Light to Metrology without Absolute Standards

Down-conversion may be used to calculate the quantum efficiency of detectors without the use of a calibrated standard. NIST have tested this and found good agreement to more conventional method. Since we essentially can use down-conversion to increase accuracy without a calibration, can we maybe use this in gate-operations in quantum computing as well.

Laboratory Exercise

Problems

Chapter 2: Field Quantization

Problem 5: Suppose that the state of a single-mode cavity is given at time $t = 0$ by

$$|\psi(0)\rangle = \frac{1}{\sqrt{2}}(|n\rangle + e^{i\varphi}|n+1\rangle) \tag{9.5}$$

where φ is some phase. Find the state $|\psi(t)\rangle$ at times $t > 0$. For this time-evolved state, verify the uncertainty relation.

Solution: We have that $|\psi(t)\rangle = \hat{U}(t)|\psi(0)\rangle$, and thus

$$\begin{aligned}
|\psi(t)\rangle &= \frac{1}{\sqrt{2}}\left(e^{-\hat{H}t/\hbar}|n\rangle + e^{i\varphi}e^{-\hat{H}t/\hbar}|n+1\rangle\right) && (9.6)\\
&= \frac{1}{\sqrt{2}}\left[\left(1 - i\frac{\hat{H}}{\hbar}t - \frac{\hat{H}^2}{\hbar^2}t^2 + i\frac{\hat{H}^3}{\hbar^3}t^3 + \cdots\right)|n\rangle\right. && (9.7)\\
&\quad \left. + e^{i\varphi}\left(1 - i\frac{\hat{H}}{\hbar}t - \frac{\hat{H}^2}{\hbar^2}t^2 + i\frac{\hat{H}^3}{\hbar^3}t^3 + \cdots\right)|n+1\rangle\right] && (9.8)\\
&= \frac{1}{\sqrt{2}}\left[\left(1 - i\omega(n+1/2)t - \omega(n+1/2)^2t^2 + i\omega(n+1/2)^3t^3 + \cdots\right)|n\rangle\right. && (9.9)\\
&\quad \left. + e^{i\varphi}\left(1 - i\omega(n+3/2)t - \omega(n+3/2)^2t^2 + i\omega(n+3/2)^3t^3 + \cdots\right)|n+1\rangle\right] && (9.10)\\
&= \frac{1}{\sqrt{2}}\left(e^{-i\omega(n+1/2)t}|n\rangle + e^{i\varphi}e^{-i\omega(n+3/2)t}|n+1\rangle\right) && (9.11)\\
&= \frac{e^{-i\omega(n+1/2)t}}{\sqrt{2}}\left(|n\rangle + e^{i(\varphi-\omega t)}|n+1\rangle\right) && (9.12)
\end{aligned}$$

Now we want to verify the uncertainty relation $\Delta\hat{X}_1\Delta\hat{X}_2 \geq 1/4$. We find that $\left\langle \hat{X}_1 \right\rangle = \cos(\varphi - \omega t)\sqrt{n+1}/2$ and $\left\langle \hat{X}_2 \right\rangle = \sin(\varphi - \omega t)\sqrt{n+1}/2$ and $\left\langle \hat{X}_1 \right\rangle = \left\langle \hat{X}_2 \right\rangle = (n+1)/2$. Calculating this yields,

$$\Delta\hat{X}_1\Delta\hat{X}_2 = \frac{n+1}{4}\sqrt{2 + \frac{\sin(2(\varphi - \omega t))}{4}} \tag{9.13}$$

which has its minimum value for $n = 0$ and $t = \varphi/\omega$, thus

$$\Delta\hat{X}_1\Delta\hat{X}_2 = \frac{1}{4}\sqrt{2} = \frac{1}{2\sqrt{2}} > \frac{1}{4}. \tag{9.14}$$

Chapter 3: Coherent States

Problem 4: Prove the following identities:

$$\hat{a}^\dagger|\alpha\rangle\langle\alpha| = \left(\alpha^* + \frac{\partial}{\partial\alpha}\right)|\alpha\rangle\langle\alpha| \quad \text{and} \quad |\alpha\rangle\langle\alpha|\hat{a} = \left(\alpha + \frac{\partial}{\partial\alpha^*}\right)|\alpha\rangle\langle\alpha|. \tag{9.15}$$

Solution: First we recall $\hat{D}(\alpha) = \exp\left(-|\alpha|^2/2\right)\exp\left(\alpha\hat{a}^\dagger\right)\exp(-\alpha^*\hat{a})$ and $\hat{D}(\alpha)\hat{a}\hat{D}^\dagger(\alpha) = \hat{a} - \alpha \iff \hat{a}^\dagger\hat{D}^\dagger(\alpha) = \hat{D}^\dagger(\hat{a}^\dagger - \alpha^*)$. Taking the derivative of the displacement operator with respect to α yields

$$\frac{\partial\hat{D}(\alpha)}{\partial\alpha} = \left(-\frac{\alpha^*}{2} + \hat{a}^\dagger\right)\hat{D}(\alpha) \quad \text{and} \quad \frac{\partial\hat{D}^\dagger(\alpha)}{\partial\alpha} = \left(-\frac{\alpha^*}{2} - \hat{a}^\dagger\right)\hat{D}^\dagger(\alpha) = \hat{D}^\dagger(\alpha)\left(\frac{\alpha^*}{2} - \hat{a}^\dagger\right). \tag{9.16}$$

Now we write

$$\frac{\partial}{\partial\alpha}|\alpha\rangle\langle\alpha| = \frac{\partial\hat{D}(\alpha)}{\partial\alpha}|0\rangle\langle\alpha| + |\alpha\rangle\langle 0|\frac{\partial\hat{D}^\dagger(\alpha)}{\partial\alpha} = \left(\hat{a}^\dagger - \frac{\alpha^*}{2}\right)|\alpha\rangle\langle\alpha| + |\alpha\rangle\langle\alpha|\left(\frac{\alpha^*}{2} - \hat{a}^\dagger\right) \tag{9.17}$$

$$= \hat{a}^\dagger|\alpha\rangle\langle\alpha| - |\alpha\rangle\langle\alpha|\hat{a}^\dagger = \hat{a}^\dagger|\alpha\rangle\langle\alpha| - \alpha^*|\alpha\rangle\langle\alpha| \implies \tag{9.18}$$

$$\hat{a}^\dagger|\alpha\rangle\langle\alpha| = \left(\alpha^* + \frac{\partial}{\partial\alpha}\right)|\alpha\rangle\langle\alpha|. \tag{9.19}$$

Recalling that $\hat{D}(\alpha)\hat{a}\hat{D}^\dagger(\alpha) = \hat{a} - \alpha \iff \hat{D}(\alpha)\hat{a} = (\hat{a} - \alpha)\hat{D}(\alpha)$, and taking the derivative of the displacement operator with respect to α^* yields

$$\frac{\partial\hat{D}(\alpha)}{\partial\alpha^*} = \hat{D}(\alpha)\left(-\frac{\alpha}{2} - \hat{a}\right) = \left(\frac{\alpha}{2} - \hat{a}\right)\hat{D}(\alpha) \quad \text{and} \quad \frac{\partial\hat{D}^\dagger(\alpha)}{\partial\alpha^*} = \hat{D}^\dagger(\alpha)\left(-\frac{\alpha}{2} + \hat{a}\right). \tag{9.20}$$

Now we write

$$\frac{\partial}{\partial\alpha^*}|\alpha\rangle\langle\alpha| = \frac{\partial\hat{D}(\alpha)}{\partial\alpha^*}|0\rangle\langle\alpha| + |\alpha\rangle\langle 0|\frac{\partial\hat{D}^\dagger(\alpha)}{\partial\alpha^*} = \left(\frac{\alpha}{2} - \hat{a}\right)|\alpha\rangle\langle\alpha| + |\alpha\rangle\langle\alpha|\left(-\frac{\alpha}{2} + \hat{a}\right) \tag{9.21}$$

$$= -\hat{a}|\alpha\rangle\langle\alpha| + |\alpha\rangle\langle\alpha|\hat{a} = -\alpha|\alpha\rangle\langle\alpha| + |\alpha\rangle\langle\alpha|\hat{a} \implies \tag{9.22}$$

$$|\alpha\rangle\langle\alpha|\hat{a} = \left(\alpha + \frac{\partial}{\partial\alpha^*}\right)|\alpha\rangle\langle\alpha|. \quad \square \tag{9.23}$$

Chapter 4: Emission and Absorption of Radiation by Atoms

Problem 6: Consider the resonant Jaynes-Cummings model for the initial thermal state as in Section 4.7. Assume the atom is initially in the excited state. Analyze the collapse of the Rabi oscillations and determine the dependence of the collapse time on the average photon number of the thermal field.

Solution: We have the inversion $W(t) = \sum_n^\infty P_n\cos(\Omega_n t)$, with $P_n = \bar{n}^n/(1+\bar{n})^{n+1}$ and $\Omega_n = 2\lambda\sqrt{n+1}$. For a given n the phase is $\varphi_n(t) = \Omega_n t = 2\lambda\sqrt{n+1}$. The spread in phase depend on the dephasing $\Delta\varphi$. This dephasing depend on the sensitivity to n, and thus we write

$$\Delta\varphi = \left.\frac{\partial\varphi_n}{\partial n}\right|_{n=\bar{n}}\Delta n = \frac{\lambda t\Delta n}{\sqrt{\bar{n}+1}} \tag{9.24}$$

and for a thermal field we have that $\Delta n \approx \bar{n}$. Thus, we get $\Delta\varphi \approx \lambda t\bar{n}/\sqrt{\bar{n}+1}$. We say that we have decoherence when the spread in phase is $\Delta\varphi \approx \pi$. This gives $t_c \approx \pi\sqrt{\bar{n}+1}/(\lambda\bar{n})$.

Chapter 5: Quantum Coherence Functions

Problem 4: Consider the superposition state of the vacuum and one-photon states, $|\psi\rangle = C_0|0\rangle + C_1|1\rangle$, where $|C_0|^2 + |C_1|^2 = 1$, and investigate its coherence properties. Note that quantum mechanically it is coherent because it is a pure state. Compare your result with that obtained for the mixture, $\hat{\rho} = |C_0|^2|0\rangle\langle 0| + |C_1|^2|1\rangle\langle 1|$.

Solution: We use $E^+ = iK\hat{a}\exp[i(\mathbf{k}\cdot\mathbf{r} - \omega t)]$ and $E^- = E^{+\dagger}$. For the superposition state we find that $G^{(1)}(x_i, x_i) = K^2|C_1|^2$ for $i = 1, 2$ and $G^{(1)}(x_1, x_2) = K^2|C_1|^2\exp[i(k_2x_2 - k_1x_1 - \omega(t_2 - t_1))]$, and clearly $g^{(1)}(x_1, x_2) = \exp[i(k_2x_2 - k_1x_1 - \omega(t_2 - t_1))]$, and thus $|g^{(2)}| = 1$. Since the state is a superposition with the biggest number state $|1\rangle$, the second order correlation function is $G^{(2)}(x_1, x_2; x_2, x_1) = 0$, since we have $\hat{a}\hat{a}$ acting on $|0\rangle$ and $|1\rangle$. This gives $|g^{(2)}(x_1, x_2; x_2, x_1)| = 0$. Doing this for the statistical mixture we find the exact same coherence functions.

Chapter 6: Beam Splitters and Interferometers

Problem 5: Let the input state to a beam splitter of arbitrary ϑ and φ be $|4\rangle_0|0\rangle_1$. Determine the corresponding output state.

Solution: We write the inverse of the scattering matrix

$$\mathbf{B}^{-1} = \frac{1}{\cos(\vartheta)} \begin{pmatrix} \cos(\vartheta/2) & \exp(i\varphi)\sin(\vartheta/2) \\ -\exp(-i\varphi)\sin(\vartheta/2) & \cos(\vartheta/2) \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} \hat{a}_0 \\ \hat{a}_1 \end{pmatrix} = \mathbf{B}^{-1} \begin{pmatrix} \hat{a}_2 \\ \hat{a}_3 \end{pmatrix}. \tag{9.25}$$

With this we can write $\hat{a}_0^\dagger = [\hat{a}_2^\dagger \cos(\vartheta/2) - \hat{a}_3^\dagger \exp(-i\varphi)\sin(\vartheta/2)]/\cos(\vartheta)$. Then, we have that the incoming state is $|4\rangle_0 |0\rangle_1 = (\hat{a}_0^\dagger)^4 |0\rangle_0 |0\rangle_1$. Carrying out the binomial calculation gives

$$|4\rangle_0 |0\rangle_1 \xrightarrow{\text{Beam splitter}} \frac{1}{\cos(\vartheta)^4} \big[\cos(\vartheta/2)^4 |4\rangle_2 |0\rangle_3 - 4\cos(\vartheta/2)^3 \exp(-i\varphi)\sin(\vartheta/2) |3\rangle_2 |1\rangle_3 \tag{9.26}$$

$$+ 6\cos(\vartheta/2)^2 \exp(-i\varphi)^2 \sin(\vartheta/2)^2 |2\rangle_2 |2\rangle_3 \tag{9.27}$$

$$- 4\cos(\vartheta/2)\exp(-i\varphi)^3 \sin(\vartheta/2)^3 |1\rangle_2 |3\rangle_3 \tag{9.28}$$

$$+ \exp(-i\varphi)^4 \sin(\vartheta/2)^4 |0\rangle_2 |4\rangle_3 \big] \tag{9.29}$$

Chapter 7: Nonclassical Light

Chapter 9: Optical Test of Quantum Mechanics